

This manuscript evaluates radiomics-based machine learning and transfer learning-based DL for predicting treatment outcome (cure versus relapse) T2-weighted MRI in GL261 glioblastoma model. The authors compare an XGBoost classifier based on handcrafted radiomic features with EfficientNetB0-based approaches and report improved discrimination and sensitivity for the deep-learning models.

The manuscript addresses an appropriate scientific question and contains several positive methodological elements, particularly the use of GLOO CVs evaluation to avoid placing examinations from the same animal in both training and test sets. However, I have several concerns regarding the definition of the independent sample, the implementation of preprocessing/model selection within CVs, the unit at which performance metrics are calculated, and the validity of some of the conclusions.

Recommendation: **Major revision.**

Major comments:

1. Would you be able to clarify if the model development pipeline was performed within the training portion of each subject level cross-val fold?

- I do like that the author used GLOO CV with all exams from one mouse withheld together. That said, in the radiomics-based pipeline, the manuscript did not sufficiently specify if their statistical analysis and feature selection steps (correlation filtering, Mann-Whitney U test step, temporal feature engineering) were refitted using only the training animals within GLOO folds. Please explicitly clarify this, especially for the feature selection step. If any of these preprocessing was performed before GLOO CVs splits, I recommend that the analysis be repeated using a fully nested subject-level procedure.

2. The author should explicitly clarify if metrics in table 1 were calculated at the level of individual MRI exams or at animal level.

- The manuscript stated that “EfficientNet correctly identified 90.9% of cured animals compared with 51.5% for XGBoost”, but these numbers does not look like they represent subject-level sensitivity. Please explicitly define the unit used for every reported performance metric. If the current metrics are examination-level, the wording should be corrected accordingly. I strongly recommend report subject-level performance because that is a more interesting than exam-level performance.

3. The manuscript overstates the evidence for early prediction performance without actually showing it.

- The temporal analysis shows that errors vary with study day, but the main AUC is calculated across examinations from the full follow-up period. Therefore, the reported AUC does not

establish predictive performance specifically during early treatment. The authors should either report performance within clearly defined early treatment windows using the existing data or revise the conclusions to avoid claiming good early prediction performance.

4. Several conclusions should be rewritten to match the reported results more precisely

- Radiomics showed higher Specificity and PPV, so DL model did not outperformed radiomics approach across all metrics as stated in the abstract. It should be “most” or the majority of.
- You chose AUC to be the main metric, but EfficientNet FE + XGB has higher AUC than FT. I do agree that the FT model is overall giving more balanced classification performance.

5. Some important tables are missing?

- The authors talked about table S3/S4 for hyperparameters search but I didn't see it in the supplementary files.

Minor comments

1. Please provide measures of uncertainty (e.g., 95% confidence intervals) for the performance metrics. Uncertainty estimation should account for clustering of examinations within animals
2. The paper acknowledges the small cohort. I recommend explicitly acknowledge that the DL did not model longitudinal sequences. The current strategy therefore uses longitudinally acquired data but is not itself a temporal sequence model.
3. PPV is pretty low across models. This should be acknowledged because it's an important point clinically still.